1-G-Coin Problem

Started on	Thursday, 17 September 2026, 12:22 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:23 PM
Time taken	32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int v, count = 0;
    int coins[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};
    int n = 9;

    if (scanf("%d", &v) != 1) return 0;

    for (int i = 0; i < n; i++) {
        while (v >= coins[i]) {
            v -= coins[i];
            count++;
        }
    }

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	49	5	5	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

2-G-Cookies Problem

Started on	Thursday, 17 September 2026, 12:23 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:23 PM
Time taken	26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input:

3

1 2 3

2

1 1

Output:

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

Constraints:

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    long long diff = (*(int*)a) - (*(int*)b);
    if (diff < 0) return -1;
    if (diff > 0) return 1;
    return 0;
}

int main() {
    int n, m;
    if (scanf("%d", &n) != 1) return 0;

    int *g = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &g[i]);
    }

    if (scanf("%d", &m) != 1) return 0;

    int *s = (int *)malloc(m * sizeof(int));
    for (int i = 0; i < m; i++) {
        scanf("%d", &s[i]);
    }

    qsort(g, n, sizeof(int), compare);
    qsort(s, m, sizeof(int), compare);

    int i = 0, j = 0;
    while (i < n && j < m) {
        if (s[j] >= g[i]) {
            i++;
        }
        j++;
    }

    printf("%d", i);

    free(g);
    free(s);
    return 0;
}
```

	Input	Expected	Got	
✔	2 1 2 3 1 2 3	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

3-G-Burger Problem

Started on	Sunday, 18 October 2026, 7:45 PM
State	Finished
Completed on	Sunday, 18 October 2026, 7:49 PM
Time taken	4 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs tr
If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calor
burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) +$
But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine
he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach t

Input Format

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

```
3
5 10 7
```

Sample Output

```
76
```

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    return (*(int *)b - *(int *)a); // Descending order
}

int main() {
    int n;
    scanf("%d", &n);

    int calories[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &calories[i]);
    }

    // Greedy: eat burgers in decreasing order of calories
    qsort(calories, n, sizeof(int), compare);

    long long total = 0;
    long long power = 1; // 3^0

    for (int i = 0; i < n; i++) {
        total += calories[i] * power;
        power *= n;
    }

    printf("%lld\n", total);

    return 0;
}
```

	Test	Input	Expected	Got	
✔	Test Case 1	3 1 3 2	18	18	✔
✔	Test Case 2	4 7 4 9 6	389	389	✔
✔	Test Case 3	3 5 10 7	76	76	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

4-G-Array Sum max problem

Started on	Thursday, 17 September 2026, 12:30 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:31 PM
Time taken	43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    long long valA = *(const long long *)a;
    long long valB = *(const long long *)b;
    if (valA > valB) return 1;
    if (valA < valB) return -1;
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;

    long long *arr = (long long *)malloc(n * sizeof(long long));
    for (int i = 0; i < n; i++) {
        scanf("%lld", &arr[i]);
    }

    qsort(arr, n, sizeof(long long), compare);

    long long max_sum = 0;
    for (int i = 0; i < n; i++) {
        max_sum += arr[i] * i;
    }

    printf("%lld", max_sum);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

5-G-Product of Array elements-Minimum

Started on	Thursday, 17 September 2026, 12:33 PM
State	Finished
Completed on	Thursday, 17 September 2026, 12:34 PM
Time taken	1 min 16 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare_asc(const void *a, const void *b) {
    long long valA = *(const long long *)a;
    long long valB = *(const long long *)b;
    if (valA > valB) return 1;
    if (valA < valB) return -1;
    return 0;
}

int compare_desc(const void *a, const void *b) {
    long long valA = *(const long long *)a;
    long long valB = *(const long long *)b;
    if (valA < valB) return 1;
    if (valA > valB) return -1;
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;

    long long *a = (long long *)malloc(n * sizeof(long long));
    long long *b = (long long *)malloc(n * sizeof(long long));

    for (int i = 0; i < n; i++) {
        scanf("%lld", &a[i]);
    }

    for (int i = 0; i < n; i++) {
        scanf("%lld", &b[i]);
    }

    qsort(a, n, sizeof(long long), compare_asc);
    qsort(b, n, sizeof(long long), compare_desc);

    long long min_sum = 0;
    for (int i = 0; i < n; i++) {
        min_sum += a[i] * b[i];
    }

    printf("%lld", min_sum);

    free(a);
    free(b);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course